Forest Adventure

Serzhan Karishal | Individual student project | 5 November 2026

Repository: github.com/sezhadlaw/ForestAdventure_GameDev

01 / Concept and player experience

A single-player 2D mobile platformer in which a forest hero crosses platforms, collects required coins, avoids hazards and reaches the level exit. The intended experience is a short, readable challenge with responsive movement and a clear goal.

Engine: Unity. Target platform: Android phones and tablets. Orientation: landscape. Visual direction: pixel-art characters and forest environments. The Unity version is recorded in ProjectSettings/ProjectVersion.txt.

02 / Core loop

Enter a level > explore and jump > collect coins > avoid or fight enemies > reach the exit > unlock the next level. The exit requires the level coin target. Damage removes health; health pickups restore it. Losing all health leads to a retry. Completing the final level leads to a win state.

03 / Controls and feedback

Left thumb: movement joystick. Right thumb: Jump and, when combat is enabled, Attack. Hearts communicate health; the coin counter shows progress toward the exit requirement. Damage, collection and button presses receive clear visual feedback; sound is part of the delivery plan.

04 / Intended content

One playable hero, three short levels, a patrolling enemy type and a final encounter. Level objects include coins, health pickups, spikes, moving platforms and exits. Difficulty grows through platform spacing, hazard combinations and enemy placement. Exact tuning follows playtesting.

This document specifies the design and acceptance targets. Planned features and tests are not presented as completed implementation.

Three screens and adaptive UI

Screen	Content and required behaviour
Main menu	Game title, Play and Quit. Play opens Level1. Keep the primary action visually dominant.
Gameplay / HUD	Joystick, Jump, Attack when available, hearts, coin counter and Pause. The play area remains readable behind the controls.
Pause overlay	Resume, Restart and Menu. Gameplay simulation pauses. Resume restores play, Restart reloads the level and Menu returns to the main menu.

05 / Scaler, anchors and safe area - 30%

Use Canvas Scaler with Scale With Screen Size; proposed reference resolution: 1920 × 1080 and Match Width Or Height: 0.5 as the starting point. Adjust after testing rather than stretching the UI to one device.

Anchor the joystick bottom-left, action buttons bottom-right, health top-left and pause top-right. Keep menu and pause groups centred using layouts. Place interactive content in a safe-area container derived from Screen.safeArea and refresh it when the display dimensions change.

Background art may extend to the display edges. Buttons, hearts, counters and text must remain inside the safe area. All three screens must fit every target device without clipping, overlap or distorted proportions.

06 / Thumbs and readability - 15%

Design targets: touch areas at least 48 × 48 density-independent pixels, preferably larger for Jump; at least 8 dp separation between adjacent controls. Verify the physical size on an Android device because Canvas units alone do not establish dp.

Place movement and action controls in comfortable lower thumb zones. Keep Pause separate from rapid action buttons. Use short labels, strong text/background contrast and visible pressed states. Review text at normal device size; avoid relying on zoomed screenshots.

07 / Visual balance

Use a consistent forest palette, sprite scale, button treatment and typography across all screens. Keep decorative detail away from labels. Use a clear hierarchy: title and primary action, supporting controls, then decoration. Avoid stretching pixel art or mixing unrelated UI styles.

Scope, ownership and production plan

08 / MVP - delivery target, week 15

A playable Android version with a main menu, responsive gameplay HUD and functional pause menu; three completable short levels; movement, jumping, coins, hazards, health, enemy encounters and a final win state. The target includes basic audio and saved progression. All three screens must fit the three device classes.

Priority order: first a complete playable loop and the assessed UI; then remaining levels and encounters; finally audio, saved progress and polish. If time becomes limited, simplify encounter complexity and decorative effects before compromising the three-screen interface or level completion.

09 / Cut list

Excluded: multiplayer, online accounts, open world, crafting, skill trees, procedural levels, in-app purchases, extra playable heroes and large enemy rosters. Additional biomes, elaborate boss phases and cosmetic effects are deferred until the assessed MVP is stable.

10 / Roles

Serzhan Karishal is responsible for game design, Unity programming, level assembly, UI layout, device testing, documentation and GitHub delivery. Third-party art and audio remain the work of their respective creators. AI assistance supports drafting and troubleshooting; the student reviews changes and verifies the game.

11 / Weekly plan

Week	Planned outcome
9	Concept, scope, repository and GDD.
10	Main menu, HUD and pause; scaler, anchors, safe area; initial device checks.
11	Enemy patrol and combat; damage and final-encounter tuning.
12	Saved progression and basic music / sound integration.
13	Finish level flow; balance; Android checks and defect fixes.
14	Run the complete 3 × 3 UI test matrix; repair layout and touch issues; collect evidence.
15	Regression test, final GDD and asset credits; verify repository access and submit links / demonstration.

Verification and evidence

12 / Device test matrix

Use the teacher-specified devices where available. Otherwise select a 16:9 phone, a phone with a notch and a 4:3 tablet in Device Simulator, all in landscape. Record the actual model and resolution in the evidence. Simulator checks establish layout coverage; physical-device testing is needed to judge touch comfort.

Device class	Main menu	Gameplay HUD	Pause
16:9 phone	Fit + Play	Fit + controls	Fit + transitions
Notch phone	Safe area + Play	Safe area + controls	Safe area + transitions
4:3 tablet	Fit + Play	Fit + controls	Fit + transitions

13 / Acceptance checks

For every matrix cell: no clipped labels, overlaps or off-screen controls; consistent proportions; readable text; safe-area compliance; and working actions. Verify Play, movement, Jump, Attack if enabled, Pause, Resume, Restart and Menu. Confirm that pausing stops gameplay and resuming restores normal time.

Gameplay checks: coins update the counter, hazards reduce health, pickups restore health within the maximum, the exit enforces its requirement, loss allows retry, and the final level reaches the win state. Planned audio and persistence require separate checks once implemented.

14 / Evidence package

Store screenshots in Docs/UI_Testing. Nine images are recommended: three screens for each device. Use descriptive filenames such as Phone16x9_MainMenu.png, NotchPhone_Gameplay.png and Tablet4x3_Pause.png. Include the simulator device name or provide a caption with model and resolution.

Record a short demonstration showing the three devices, screen transitions and key controls. Host a large video separately and link it from README. Screenshots demonstrate layout; video demonstrates interactions. Neither substitutes for a test result log.

15 / Review status

Repository inspection confirmed that nine screenshots were uploaded. Their presence does not certify that every visual or interaction check passed. Record actual pass/fail results, defects and retests after running the matrix; do not mark unperformed checks as passed.

Repository delivery and assessment

16 / Git workflow and access - 20%

Deliver Assets, Packages, ProjectSettings, README and Docs at repository root. Use the GitHub Unity .gitignore. Keep Library, Temp, Logs, UserSettings and build outputs outside version control. Preserve resource .meta files; use Visible Meta Files and Force Text.

The repository must be public or the teacher must be invited to a private repository. Keep at least three meaningful commits from the student account and at least one feature branch merged into main through a pull request. Describe actual changes rather than creating empty commits to increase the count.

At review, the history included project import, README documentation and UI screenshots, with a documentation branch merged through PR #1. Repository status can change; verify access and the main branch again before submission.

17 / Resource credits and licensing

Forest background resources are stored under Assets/Art/ForestBackground. The repository includes a public-license.pdf within the imported forest pack. Preserve license documents. The original draft did not establish the asset author or download URL, so attribution must be completed from the original source or license rather than guessed.

For each external art, font and sound asset, record: asset title, creator, original source URL, license and any modifications. Reuse only under the applicable license. Existing folders alone do not establish authorship or usage rights.

18 / Assessment mapping

Criterion	Evidence in this document / delivery
Adaptive screens - 30%	Sections 05 and 12-15; three screens across three devices.
GDD - 25%	Core loop (02), MVP (08), cut list (09), roles (10), plan (11).
Git - 20%	Section 16; meaningful commits, merged PR and clean accessible repository.
Thumbs / readability - 15%	Section 06; target sizes, placement, spacing and text checks.
Visual balance	Section 07; consistent forest style and hierarchy. Weight was not specified in the supplied criteria.

Submission blockers: Library or build files in the repository, UI that fits only one screen size, or a repository the teacher cannot open. Remaining review items: actual UI test results, final implementation checks and verified asset credits.